

## Unit 116 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 89.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 44 plots, 507 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

**Table 1. Stand summary from the BAF 20 cruise, per acre**

| STAND DENSITY        |                                       | STAND INFO                  |                                                           |
|----------------------|---------------------------------------|-----------------------------|-----------------------------------------------------------|
| Basal area           | 230 sq ft/ac (1 SE of mean, BA 1.5 %) | Type                        | Sierran mixed conifer                                     |
| Trees per acre       | 70                                    | CWHR size / density         | 5 / D (cover 94 %)                                        |
| QMD                  | 24.6 in                               | SDI                         | 283 (49 % of max 573)                                     |
| SAWTIMBER (>= 10 in) |                                       | BIOMASS (< 10 in)           |                                                           |
| BA   QMD             | 230 sq ft/ac   24.9 in                | BA   QMD                    | 1 sq ft/ac   9.4 in                                       |
| TPA                  | 68                                    | TPA                         | 2                                                         |
| Volume               | 40.1 MBF/ac (7993 cu ft)              | Volume                      | 0.5 green tons/ac (19 cu ft)                              |
| TREATMENT TARGET     |                                       | CRUISE DESIGN               |                                                           |
| Leave                | 35 % of max SDI = ~163 sq ft/ac BA    | Sampling error (volume)     | 4.5 % at 95 % (CV 15 %, t = 2.02, n = 44)                 |
| Species (BA)         | PP 65, WF 62, DF 42, IC 36, SP 25     | Stratum standard            | 40 % of volume - MEETS; plots for standard: 1             |
|                      |                                       | Sampling error (basal area) | 1 % at 95 % (CV 10 %); secondary, not the standard        |
|                      |                                       | Plot count                  | 44 vs 20-plot local practice, not a handbook standard: ok |

Cubic volume = total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium) net of defect; Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006); green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber, so applied to total-stem CVTS it overstates sawlog volume; merchantable cubic volume would be the consistent pairing. Sampling error at 95 % confidence with Student's t on plot net cubic volume per acre, the quantity the handbook standards apply to (FSH 2409.12 ch. 40, 41.1(5)(a) and (b)); stratum standard 40 % of volume for tree-measurement sales; the basal-area error is secondary. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables exclude records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise\_data.xlsx, Standards and assumptions.

**Table 2. Stand table by 4-inch DBH class, per acre**

| DBH class, in | Trees      | TPA         | BA           | cu ft       | MBF         |
|---------------|------------|-------------|--------------|-------------|-------------|
| 8-11.9        | 6          | 4.9         | 2.7          | 54          | 0.3         |
| 12-15.9       | 27         | 10.5        | 12.3         | 294         | 1.5         |
| 16-19.9       | 57         | 14.7        | 25.9         | 689         | 3.5         |
| 20-23.9       | 71         | 12.3        | 32.3         | 946         | 4.7         |
| 24-27.9       | 86         | 10.7        | 39.1         | 1295        | 6.5         |
| 28-31.9       | 70         | 6.4         | 31.8         | 1147        | 5.8         |
| 32-35.9       | 58         | 4.3         | 26.4         | 1002        | 5.0         |
| 36-39.9       | 44         | 2.6         | 20.0         | 829         | 4.2         |
| 40-43.9       | 29         | 1.4         | 13.2         | 546         | 2.7         |
| 44-47.9       | 15         | 0.6         | 6.8          | 287         | 1.4         |
| 48-51.9       | 21         | 0.7         | 9.5          | 422         | 2.1         |
| 52-55.9       | 10         | 0.3         | 4.5          | 214         | 1.1         |
| 56-59.9       | 5          | 0.1         | 2.3          | 112         | 0.6         |
| 60-63.9       | 8          | 0.2         | 3.6          | 175         | 0.9         |
| <b>All</b>    | <b>507</b> | <b>69.7</b> | <b>230.5</b> | <b>8012</b> | <b>40.2</b> |

**Table 3. Stock table by species, per acre**

| Species    | Trees      | TPA         | BA           | QMD         | cu ft       | MBF         |
|------------|------------|-------------|--------------|-------------|-------------|-------------|
| PP         | 144        | 17.6        | 65.5         | 26.1        | 2562        | 12.9        |
| WF         | 136        | 19.6        | 61.8         | 24.0        | 2143        | 10.8        |
| DF         | 93         | 13.1        | 42.3         | 24.4        | 1386        | 7.0         |
| IC         | 79         | 11.8        | 35.9         | 23.6        | 1001        | 5.0         |
| SP         | 55         | 7.6         | 25.0         | 24.6        | 919         | 4.6         |
| <b>All</b> | <b>507</b> | <b>69.7</b> | <b>230.5</b> | <b>24.6</b> | <b>8012</b> | <b>40.2</b> |

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 116: plots on the 300 ft grid (orange = QA flag)

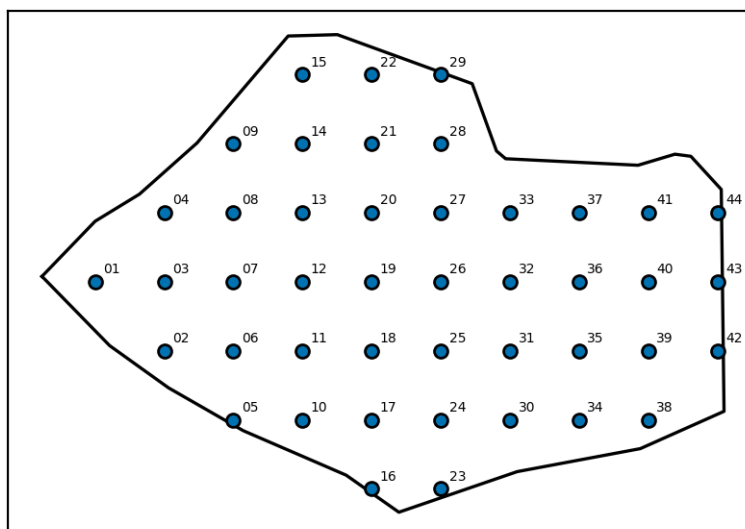


Figure 1. Plot locations with QA status.

**QA findings:** no issues found; unit accepted

Table 4. Plot list

| Plot   | Trees in | BA sq ft/ac | Slope % | Status |
|--------|----------|-------------|---------|--------|
| 116-01 | 10       | 200         | 17      | ok     |
| 116-02 | 12       | 240         | 17      | ok     |
| 116-03 | 11       | 220         | 11      | ok     |
| 116-04 | 10       | 200         | 20      | ok     |
| 116-05 | 11       | 220         | 24      | ok     |
| 116-06 | 12       | 240         | 15      | ok     |
| 116-07 | 10       | 200         | 14      | ok     |
| 116-08 | 10       | 200         | 11      | ok     |
| 116-09 | 12       | 240         | 14      | ok     |
| 116-10 | 11       | 220         | 19      | ok     |
| 116-11 | 11       | 220         | 21      | ok     |
| 116-12 | 14       | 280         | 12      | ok     |
| 116-13 | 11       | 220         | 10      | ok     |
| 116-14 | 11       | 220         | 4       | ok     |
| 116-15 | 12       | 240         | 3       | ok     |
| 116-16 | 10       | 200         | 15      | ok     |
| 116-17 | 13       | 260         | 12      | ok     |
| 116-18 | 11       | 220         | 16      | ok     |
| 116-19 | 12       | 240         | 20      | ok     |
| 116-20 | 12       | 240         | 12      | ok     |
| 116-21 | 11       | 220         | 8       | ok     |
| 116-22 | 10       | 200         | 4       | ok     |
| 116-23 | 11       | 220         | 12      | ok     |
| 116-24 | 13       | 260         | 12      | ok     |
| 116-25 | 11       | 220         | 20      | ok     |
| 116-26 | 11       | 220         | 14      | ok     |
| 116-27 | 13       | 260         | 11      | ok     |
| 116-28 | 13       | 260         | 16      | ok     |
| 116-29 | 13       | 260         | 27      | ok     |
| 116-30 | 11       | 220         | 22      | ok     |
| 116-31 | 12       | 240         | 20      | ok     |
| 116-32 | 14       | 280         | 12      | ok     |
| 116-33 | 11       | 220         | 6       | ok     |
| 116-34 | 10       | 200         | 10      | ok     |
| 116-35 | 11       | 220         | 14      | ok     |
| 116-36 | 11       | 220         | 11      | ok     |
| 116-37 | 12       | 240         | 11      | ok     |
| 116-38 | 13       | 260         | 13      | ok     |
| 116-39 | 12       | 240         | 9       | ok     |
| 116-40 | 13       | 260         | 10      | ok     |
| 116-41 | 9        | 180         | 5       | ok     |
| 116-42 | 11       | 220         | 11      | ok     |
| 116-43 | 13       | 260         | 4       | ok     |
| 116-44 | 12       | 240         | 19      | ok     |